

Question	Marking Guidance	Mark	Comments
03.1	1. Microvilli increase surface area for diffusion/facilitated diffusion/active transport/co-transport OR Microvilli increase surface area for (more) channel/carrier proteins; 2. (Many) mitochondria release energy/ATP for active transport;	2 (2 x AO2)	1. Accept folded cell (surface) membrane or brush border for microvilli 1. Accept high/large for 'increased' 1. Accept 'they' for microvilli 2. Accept 'uptake' for 'transport' 2. Accept 'cotransport' for 'active transport'

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03.2	Correct answer for 2 marks, 40–41(:1);; Accept for 1 mark, 0.32–0.33 (correct surface area of a microvillus) OR 321–322 (correct cell surface area) OR Ratio not simplified e.g. 321:8;	2 (2 x AO2)	

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03.3	1. ZO-1 is located in cell (surface) <u>membrane</u>; 2. Antibody is <u>complementary</u> (to ZO-1); 3. (So) binds/attaches to the ZO-1/protein; 4. (Cells identified with) dye/stain/fluorescent marker linked to antibody;	3 max (3 x AO2)	2. and 3. Reject 'active site' once 3. Accept 'forms antigen-antibody complex' 4. Accept attachment of coloured substance to antibody, but 'colour production' on its own is insufficient. 4. 'Marker' on its own is insufficient.